ANDROID DEVELOPMENT GUIDE

Material Design

Google's design system for building consistent, accessible, and visually appealing Android apps



Core principles
Surfaces, bold design, and meaningful motion

Components
Buttons, cards, FABs, navigation, and app bars

Also covers
Color system, typography, elevation, and theming

Level
Beginner to intermediate Android developers

Covers Material Design 2 and Material Design 3 (Material You), core components, theming, and real-world usage in Google apps.

OVERVIEW

What Is Material Design?

Material Design is a design system created by **Google** that provides guidelines, reusable components, and best practices for building attractive, consistent, and user-friendly Android applications. It helps developers create apps that are consistent in appearance, easy to use, responsive across screen sizes, accessible, and visually appealing.

> **Good to know:** Material Design launched in **2014**. Google introduced its next major revision, **Material Design 3 ("Material You")**, in **2021 alongside Android 12**. Material You's signature feature is **dynamic color** — the system extracts a color palette from the user's wallpaper at runtime (requires Android 12 / API 31 or higher) and applies it across the UI.

Objectives

Consistency, usability, accessibility, responsive layouts, and standard design principles.

Core Principles

Material as a surface, bold & intentional design, and meaningful motion.

Components

Buttons, text fields, cards, FABs, navigation drawers, bottom nav, and app bars.

Foundations

Color system, typography scale, iconography, and elevation.

1 Material as a Surface

Material Design treats UI elements like **physical surfaces or sheets of paper**. Each component has height (elevation), shadows, and clear boundaries — helping users understand which elements are clickable or layered above others.

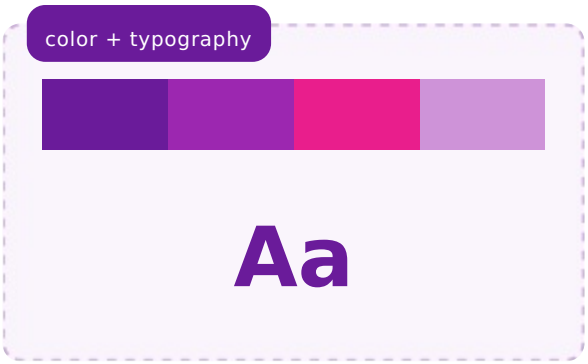



```
<com.google.android.material.card.MaterialCardView
  android:layout_width="match_parent"
  android:layout_height="wrap_content"
  app:cardElevation="4dp"
  app:cardCornerRadius="12dp"/>
```

Benefit: Higher elevation values (measured in **dp**) mean a component sits closer to the user, casting a larger, softer shadow. In Material 3, elevation is also communicated through a subtle **tonal surface tint** in addition to shadow — useful in dark theme, where shadows are far less visible.

2 Bold, Graphic & Intentional Design

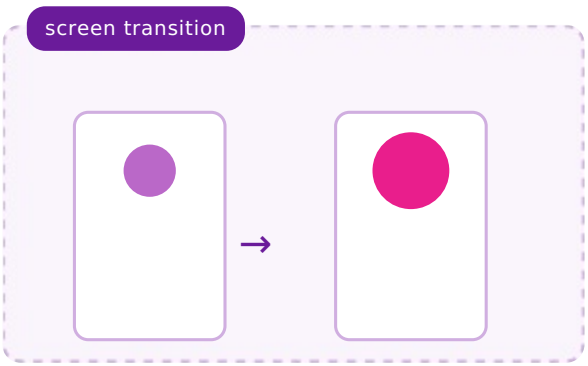
Material Design uses bright, deliberate colors, large typography, clear icons, high-quality images, and generous white space. Together these choices improve readability and give an app a modern, confident appearance.

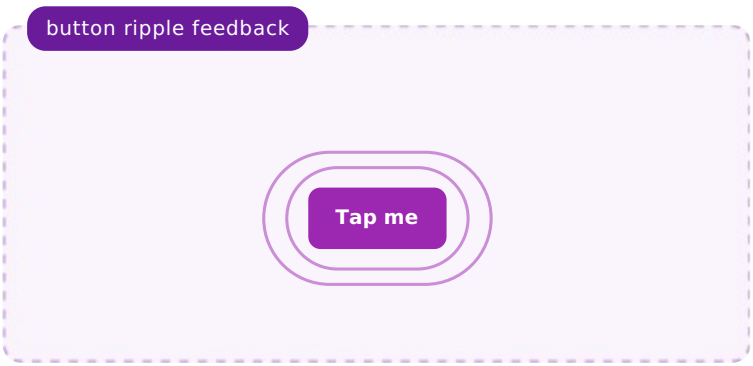


In practice: a bold headline, a confident accent color, and one strong call-to-action button do more for usability than a dozen competing visual elements. Material Design favors a small, purposeful color palette used consistently over many small variations.

3 Meaningful Motion

Animations guide user attention, show transitions between screens, and provide feedback after actions — such as a button ripple, screen transitions, loading indicators, and expand/collapse animations. Motion should be smooth and purposeful, never distracting.

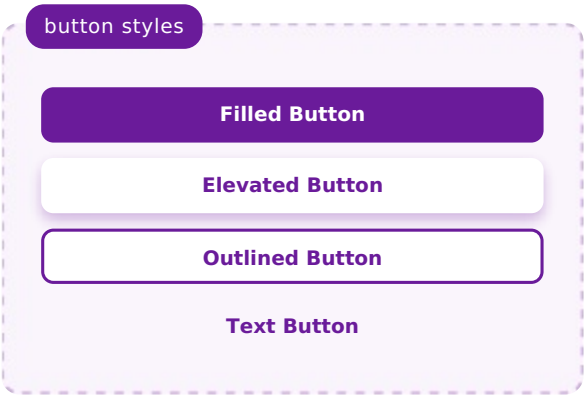




Duration matters: Material guidance recommends most UI transitions last roughly **150–300 milliseconds** — long enough to feel smooth, short enough that the interface still feels responsive.

4 Material Buttons

Buttons let users perform actions. Material Design defines four standard styles, each signaling a different level of emphasis: Filled (highest emphasis), Elevated, Outlined, and Text (lowest emphasis).

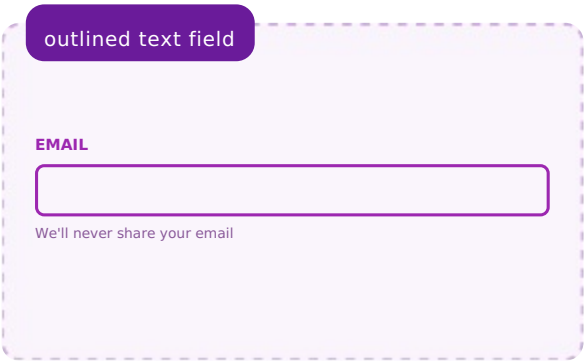



```
<com.google.android.material.button.MaterialButton
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:text="Submit"/>
```

> **Guidance:** use one **Filled** button for the single most important action on a screen; reserve **Text** buttons for low-emphasis actions like "Cancel" inside dialogs.

5 Text Fields

Text fields capture user input such as usernames, passwords, and email addresses. Material text fields support floating labels, inline error messages, and helper text.

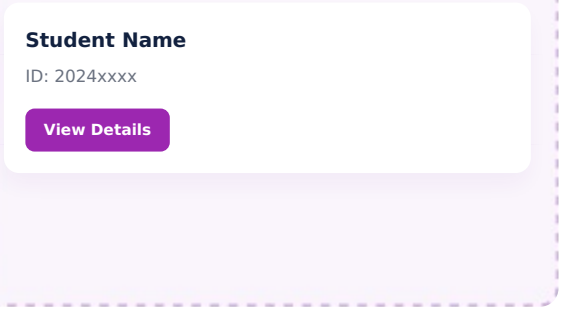


> **Three common states:** a text field can show a floating label at rest, an active state with a highlighted outline while focused, and an error state with red outline plus an inline error message when validation fails.

6 Cards

Cards group related information into a single, self-contained surface — commonly with rounded corners and a soft shadow to indicate elevation above the background.

material card




```
<MaterialCardView
  app:cardCornerRadius="12dp"
  app:cardElevation="3dp">

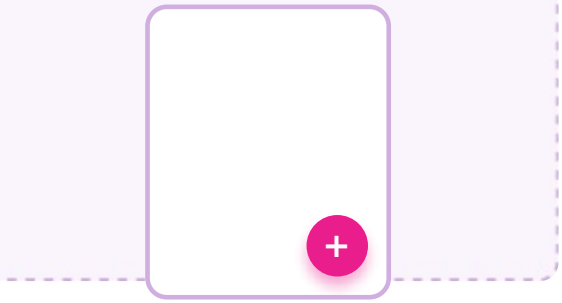
  <TextView android:text="Student Name"/>
  <Button android:text="View Details"/>

</MaterialCardView>
```

7 Floating Action Button (FAB)

A **Floating Action Button** represents the single primary action on a screen — common uses include adding a contact, composing an email, or creating a note.

FAB — primary action

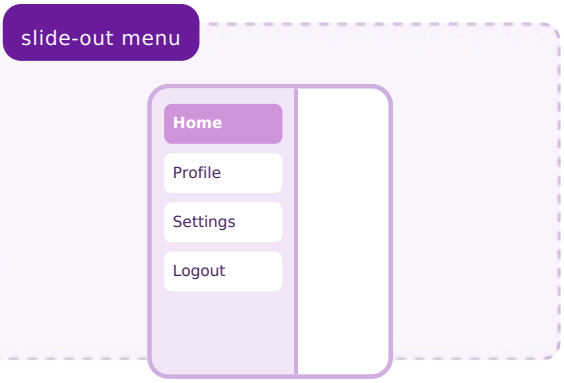


> **Best practice:** use only **one FAB per screen**, conventionally placed in the bottom-right corner (in left-to-right layouts) with a default margin of about **16dp** from the screen edges.

8 Navigation Drawer

A navigation drawer is a side menu that slides in from the left, listing an app's top-level destinations — useful for apps with many sections.

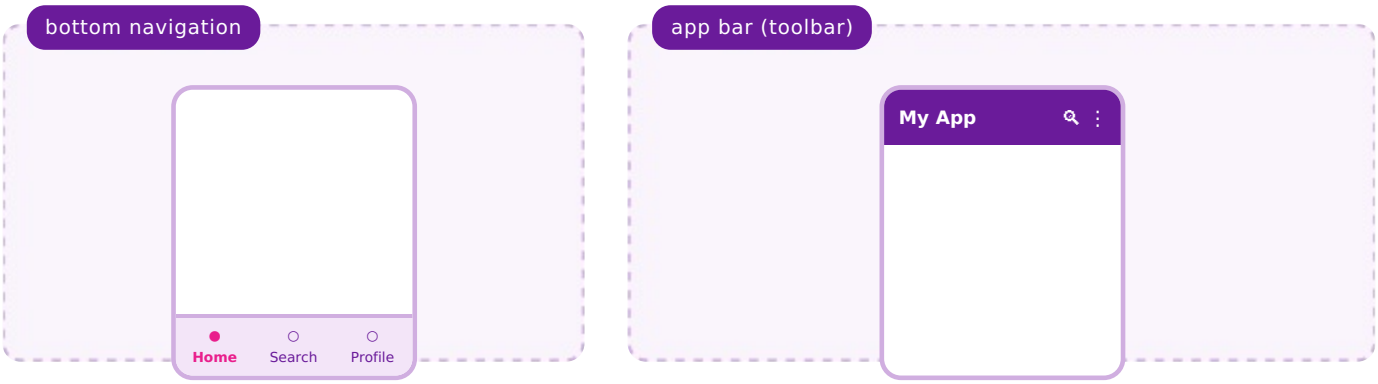
slide-out menu



> **Guidance:** Google's guidance suggests a navigation drawer works best for **5 or more** top-level destinations. For 3–5 destinations, **bottom navigation** is usually a faster, more thumb-friendly choice.

9 Bottom Navigation & App Bar

Bottom navigation switches between an app's major sections and is common in shopping, banking, and social apps. The App Bar (Toolbar) sits at the top of the screen, typically showing the app title, a navigation icon, search, and action buttons.



> **Rule of thumb:** use bottom navigation for **3–5** top-level destinations; beyond that, a navigation drawer (or a "More" tab) scales better.

10 Material Design Theme

A theme is a predefined set of styles that controls an app's overall appearance — including colors, typography, shapes, buttons, and icons — applied consistently across every screen.

> **DayNight matters:** basing a custom theme on a `DayNight` parent (as shown here) gives the app automatic light and dark mode support without writing separate theme files for each.

```
<style name="Theme.MyApp"
  parent="Theme.Material3.DayNight.NoActionBar">

  <item name="colorPrimary">
    @color/purple_700
  </item>
</style>
```

11 Material Design Color System

Material Design recommends a consistent, limited color palette built from defined roles rather than arbitrary hex values.



Primary
Main brand color

Secondary
Accent color

Background
Screen background

Surface
Cards, sheets, dialogs

Error
Validation & warnings

Tertiary
Material 3 accent role

> **Material 3 update:** rather than a single fixed hex per role, M3 defines each color as a **tonal palette** (13 tonal steps) so the same role automatically produces correct light-theme and dark-theme values, and can shift with dynamic color.

12 Typography

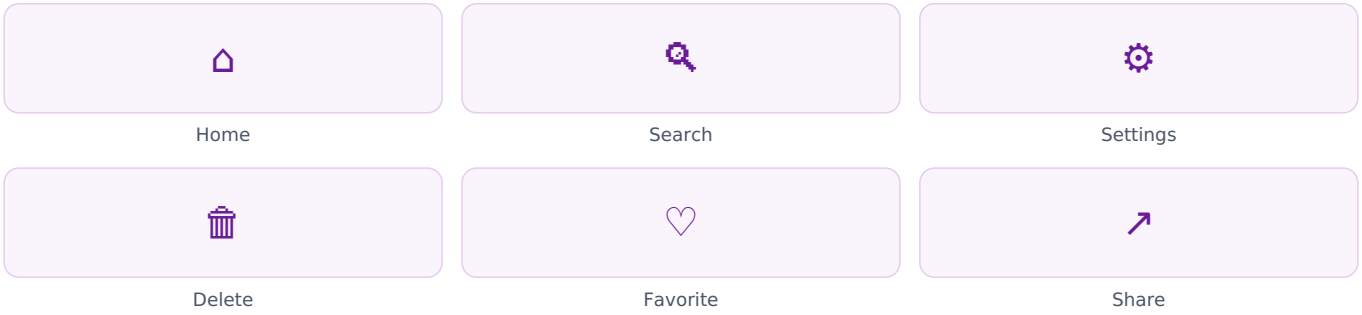
Material Design defines a type scale so text sizes stay consistent and readable across an app.

Headline	**Aa Bb Cc**
Title	**Aa Bb Cc**
Body	Aa Bb Cc
Label	**Aa Bb Cc**
Caption	Aa Bb Cc

> **Accessibility tip:** Material guidance recommends body text stay at or above roughly **14sp**, since sp units scale with the user's system font-size setting — unlike fixed dp — which keeps text readable for people who increase their device's text size.

13 Icons

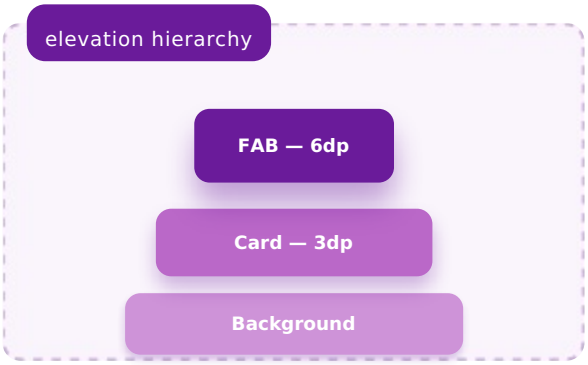
Material Design provides a large open-source library of **Material Symbols** (formerly Material Icons) that help users quickly recognize actions and navigation options.



> **Good to know:** Material Symbols ship in three styles — **Outlined**, **Rounded**, and **Sharp** — plus adjustable weight and fill, so icon style can match an app's overall visual tone.

14 Elevation and Shadows

Material Design uses elevation to indicate hierarchy — higher elevation means a component sits closer to the user and casts a more pronounced shadow.



> **Typical resting elevations:** cards commonly rest around **1–3dp**, app bars around **0–4dp** (0 when scrolled to top in M3), and a FAB around **6dp**, rising higher while being pressed or dragged.

15 Responsive Design

Material Design supports a wide range of devices — smartphones, tablets, foldables, and Chromebooks — with layouts that automatically adapt to different screen sizes and orientations.



Modern approach: Google now recommends designing against **window size classes** (Compact, Medium, Expanded) rather than individual device models, so one layout definition adapts correctly whether it's rendered on a phone or the large screen of an unfolded foldable.

16 Advantages & Disadvantages

Advantages

- Consistent, attractive, and modern user interface
- Easy navigation and responsive across devices
- Rich library of reusable, tested UI components
- Improved accessibility and faster UI development

Disadvantages

- Beginners may find the full component set challenging
- Advanced customization can get complex
- Apps using the same system may look visually similar

17 Real-World Applications

Application	Material Design Features
Gmail	Floating Action Button, Cards, Navigation Drawer
Google Drive	Cards, Bottom Navigation, Material Buttons
Google Maps	App Bar, Cards, Dynamic Color (Material You)
Google Photos	Grid Layout, Cards, Bottom Navigation

SUMMARY

Key Takeaways

Material Design is Google's recommended design system for Android applications. It provides standardized UI components, design principles, and themes that help developers create consistent, responsive, accessible, and visually appealing applications — from Material surfaces with elevation and shadows, to bold visual design, meaningful motion, reusable components, and adaptive, responsive layouts.

Exam tip: if asked to explain Material Design, anchor your answer on its three core principles: (1) **Material as a Surface** — elevation and shadows, (2) **Bold, Graphic and Intentional Design** — color, typography, and icons, and (3) **Meaningful Motion** — animations and transitions that guide attention without distracting from the task at hand.